

Formulae Sheet

Human Resources Management

Absenteeism (%)

$$\text{Absenteeism} = \frac{\text{Number of days of employee absence}}{\text{Total number of workdays}} \times 100$$

Unit 3 - Marketing

Market Share (%)

$$\text{Market Share} = \frac{\text{Sales of the business in time period}}{\text{Total market sales in time period}} \times 100$$

Price Elasticity of Demand (PED)

$$\text{PED} = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in price}}$$

Promotional Elasticity of Demand (PED)

$$\text{Promotional Elasticity of Demand} = \frac{\% \text{ Change in demand for the product}}{\% \text{ Change in promotional spending}}$$

Income elasticity of demand (IED)

$$\text{Income Elasticity} = \frac{\% \text{ Change in demand for the product}}{\% \text{ Change in consumer incomes}}$$

Median Value

$$\text{Median Value} = \frac{\text{Number of values} + 1}{2}$$

Range

$$\text{Range} = \text{Highest result} - \text{Lowest result}$$

Section Angle (Pie Graph)

$$\text{Angle of Section} = \frac{\text{Value of one section}}{\text{Total value of all sections}} \times 360 \text{ degrees}$$

Unit 4 - Operations Management

Average Cost

$$\text{Average cost} = \frac{\text{Total cost of producing this product}}{\text{Number of units produced}}$$

Break-Even

$$\text{Break-even level of output} = \frac{\text{Fixed cost}}{\text{Contribution per unit}}$$

Capacity utilisation

$$\text{Rate of capacity utilisation} = \frac{\text{Current output level}}{\text{Maximum output level}} \times 100$$

Labour Productivity

$$\text{Labour productivity (number of units per worker)} = \frac{\text{Total output in a given time period}}{\text{Total workers employed}}$$

Margin of Safety

$$\text{Margin of safety} = \text{Break-even level of output} - \text{Current output level}$$

Production Over Break-Even Point

$$\text{Production over break-even point} = \frac{\text{Margin of safety}}{\text{Break-even output}}$$

Unit Contribution

$$\text{Unit contribution} = \text{Price of unit} - \text{Direct (variable) cost of unit}$$

Total Contribution

$$\text{Total Contribution} = \text{Unit contribution} \times \text{Output}$$

Free Float

$$\text{Free Float} = \text{EST (next activity)} - \text{Duration} - \text{EST (this activity)}$$

Total Float

$$\text{Total Float} = \text{LFT} - \text{Duration} - \text{EST}$$

Unit 5 - Finance and Accounting**Closing Cash Balance**

$$\text{Closing Cash Balance} = \text{Opening cash balance} + [\text{Cash inflows} - \text{Cash outflows}]$$

Revenue

$$\text{The total value of sales made during the trading period} = \text{Selling price} \times \text{Quantity sold}$$

Cost of Sales

$$\text{Cost of sales} = (\text{Opening Inventories} + \text{Purchases}) - \text{Closing Inventories}$$

Gross Profit

$$\text{Gross Profit} = \text{Revenue} - \text{Cost of sales}$$

Gross Profit Margin Ratio (%)

$$\text{Gross Profit Margin Ratio} = \frac{\text{Gross Profit}}{\text{Revenue}} \times 100$$

Profit

$$\text{Profit} = \text{Contribution} - \text{Overheads}$$

Operating Profit Margin Ratio (%)

$$\text{Operating Profit Margin Ratio} = \frac{\text{Operating Profit (profit from operations)}}{\text{Revenue}} \times 100$$

Annual Depreciation

$$\text{Annual Depreciation} = \frac{\text{Original Cost of Asset} - \text{Expected Residual Value}}{\text{Expected useful life of asset (years)}}$$

Working Capital

$$\text{Working Capital} = \text{Current assets} - \text{Current liabilities}$$

Annual Depreciation

$$\text{Annual Depreciation} = \frac{\text{Original Cost of Asset} - \text{Expected Residual Value}}{\text{Expected useful life of asset (years)}}$$

Net Book Value

The current statement of financial position value of a non-current asset = Original cost - accumulated depreciation

Net Realisable Value

Net Realisable Value = The amount for which the existing inventory can be sold – Cost of selling it

Reserves

$$\text{Reserves} = \text{Shareholders' equity} - \text{Share capital}$$

Liquid Assets

$$\text{Liquid Assets} = \text{Current assets} - \text{Inventories}$$

Payback Period - Additional Months

$$\text{Additional Months to Payback} = \frac{\text{Additional Net Cash Flow needed}}{\text{Annual Cash Flow in a year}} \times 12 \text{ months}$$

Accounting Rate of Return (ARR)

$$\text{ARR} = \frac{\text{Average annual profit}}{\text{Average investment}} \times 100$$

Current Ratio

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Acid Test Ratio

$$\text{Acid Test Ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

Capital Employed (%)

Capital Employed = Non-current liabilities + Shareholders' equity (where shareholders' equity = Issued shares + Reserves)

Return On Capital Employed Ratio (ROCE)

$$\text{Return On Capital Employed Ratio} = \frac{\text{Operating profit}}{\text{Capital employed}} \times 100$$

Market Capitalisation

Market capitalisation = Current share price $\times$ Total number of shares issued

Average Investment

$$\text{Average Investment} = \frac{\text{Initial capital cost} + \text{Residual capital value}}{2}$$

Trade Payables Turnover (days)

$$\text{Trade Payables Turnover} = \frac{\text{Trade Payables}}{\text{Credit Purchases}} \times 365$$

Trade Receivables (days)

$$\text{Trade Receivables Turnover} = \frac{\text{Trade Receivables}}{\text{Credit Sales}} \times 365$$

Average Inventory

$$\text{Average inventory} = \frac{\text{Inventory at start of year} + \text{Inventory at end of year}}{2}$$

Rate of Inventory Turnover

$$\text{Rate of Inventory Turnover} = \frac{\text{Cost of Sales}}{\text{Average Inventory}}$$

Gearing Ratio (%)

$$\text{Gearing ratio} = \frac{\text{Non-current liabilities}}{\text{Capital employed}} \times 100$$

Dividend Cover Ratio (x)

$$\text{Dividend cover ratio} = \frac{\text{Profit for the year}}{\text{Annual dividends}}$$

Dividend per Share

$$\text{Dividend per share} = \frac{\text{Total annual dividends}}{\text{Total number of issued shares}}$$

Dividend Yield Ratio (%)

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend per share}}{\text{Market share price}} \times 100$$

Earnings per share (EPS)

$$\text{Earnings Per Share} = \frac{\text{Profit for the year}}{\text{Number of shares issued}}$$

Price/Earnings (P/E) Ratio

$$\text{P/E Ratio} = \frac{\text{Market Share Price}}{\text{Earnings per Share}}$$
